A-3.2 - Pre-Installation Procedures

This section describes how to prepare for installing both the FastBack® Model 90E (the 4-footed drive unit) and the Model 90E-G2 (3-footed drive unit). The model 90E-G2 is the most current version as of the writing of this manual. The following topics are discussed in this section:

- Foot Hole Pattern (90E only)
- Load Data (90E only)
- Foot Hole Pattern (90E-G2 only)
- Load Data (90E-G2 only)
- Structural Requirements (90E and 90E-G2)
- Electrical Requirements(90E and 90E-G2)

Foot Hole Pattern (90E only)

Figure 3-1 shows the foot hole pattern for the FastBack® Model 90E. Figure 3-2 shows the detail for the articulating foot.

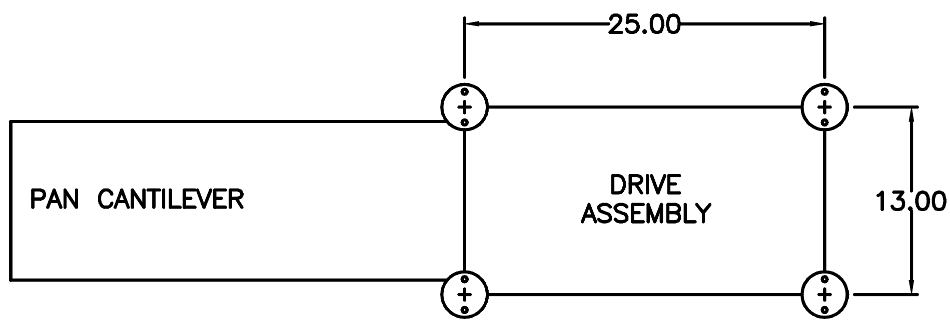

Figure 3-1 – Model 90E Foot Hole Pattern

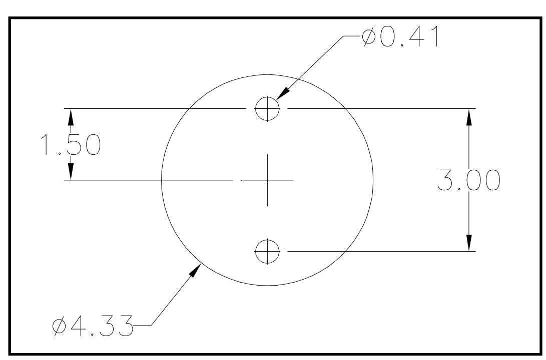

Figure 3-2 – Model 90E Articulating Foot Detail

Load Data (90E only)

Table 3-1 provides load data for the FastBack® Model 90E. Figure 3-3 shows the load point locations for a FastBack® Model 90E with a maximum-weight pan.

Static Vertical Load Per Articulating Foot (Pounds)		**Total Static Vertical Load (Pounds)**	**Dynamic Vertical Loading**		**Dynamic Horizontal Loading (Forward/Aft)**	
A	**B**		**Pounds**	**Freq. (Hz)**	**Pounds**	**Freq. (Hz)**
105	90	390	±90	3.0	±27	3.0

Table 3-1 – FastBack® Model 90E Load Data

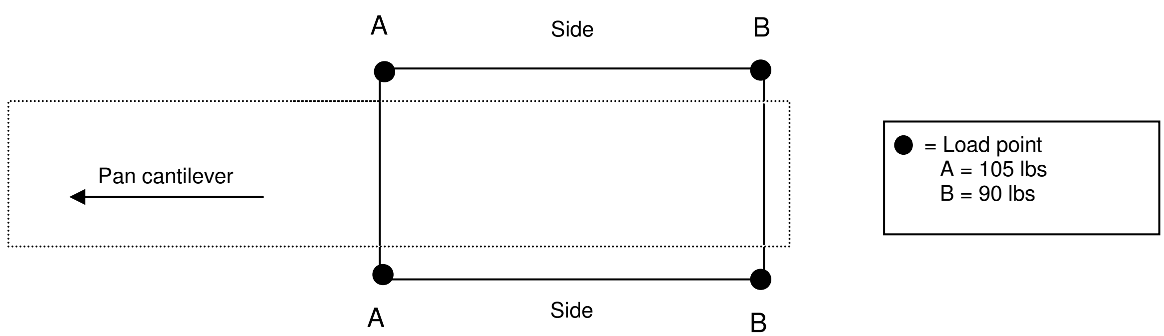

Figure 3-3 – Model 90E Load Point Locations

Foot Hole Pattern (90E-G2 only)

Figure 3-4 shows the foot hole pattern for the FastBack® Model 90E-G2. Figure 3-5 shows the detail for the articulating foot.

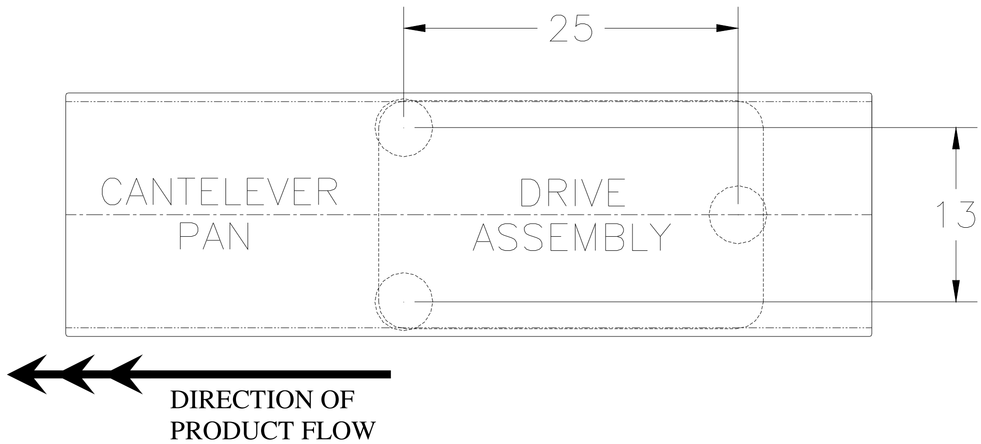

Figure 3-4 – Model 90E-G2 Foot Hole Pattern

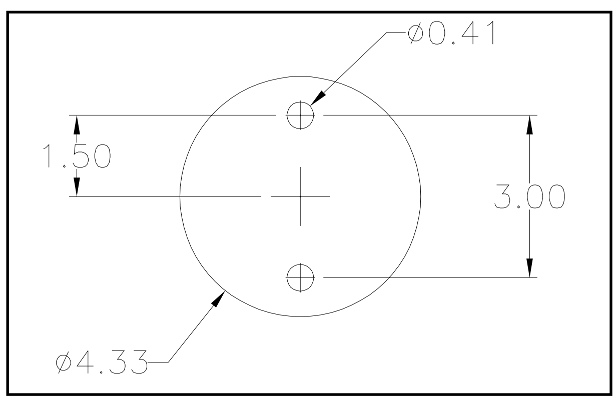

Figure 3-5 – Model 90E-G2 Articulating Foot Detail

Load Data (90E-G2 only)

Table 3-2 provides load data for the FastBack® Model 90E-G2. Figure 3-6 shows the load point locations for a FastBack® Model 90E-G2 with a maximum-weight pan.

Static Vertical Load Per Articulating Foot (Pounds)		**Total Static Vertical Load (Pounds)**	**Dynamic Vertical Loading**		**Dynamic Horizontal Loading (Forward/Aft)**	
A	**B**		**Load at Point A (Pounds)**	**Freq. (Hz)**	**Load at Point A (Pounds)**	**Freq. (Hz)**
105	180	390	±105	3.0	±27	3.0

Note: Dynamic Load Points for the single-foot location (point B) is approximately double that of the double-foot location (Point A). The Frequencies are the same for each location.

Table 3-2 – FastBack® Model 90E-G2 Load Data

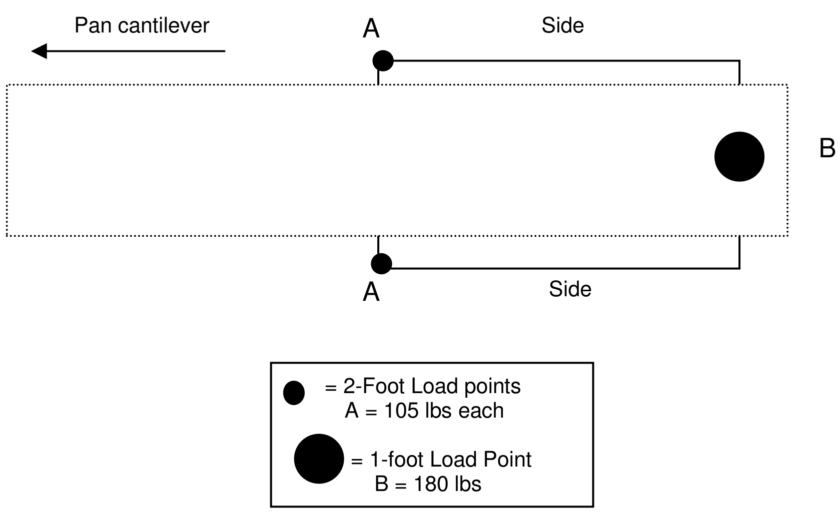

Figure 3-6 – Model 90E-G2 Load Point Locations

Structural Requirements (90E and 90E-G2)

Heat and Control provides engineering services to design conveyor support platforms for the FastBack® Models 90E and 90E-G2. Below are general guidelines for designing an appropriate support platform.

- Every distribution line that includes a FastBack® Model 90E or 90E-G2 should have a central spine, consisting of 6x6x¼-inch tubular steel, that runs down the center of the line, especially when T-pedestals are used to elevate the conveyor. T-pedestals should be welded to the 6x6x¼-inch central spine as shown in Figure 3-7. When the FastBack® arrives, the articulating feet should be bolted to the T-pedestals (see Section 4 – Installation Procedures).

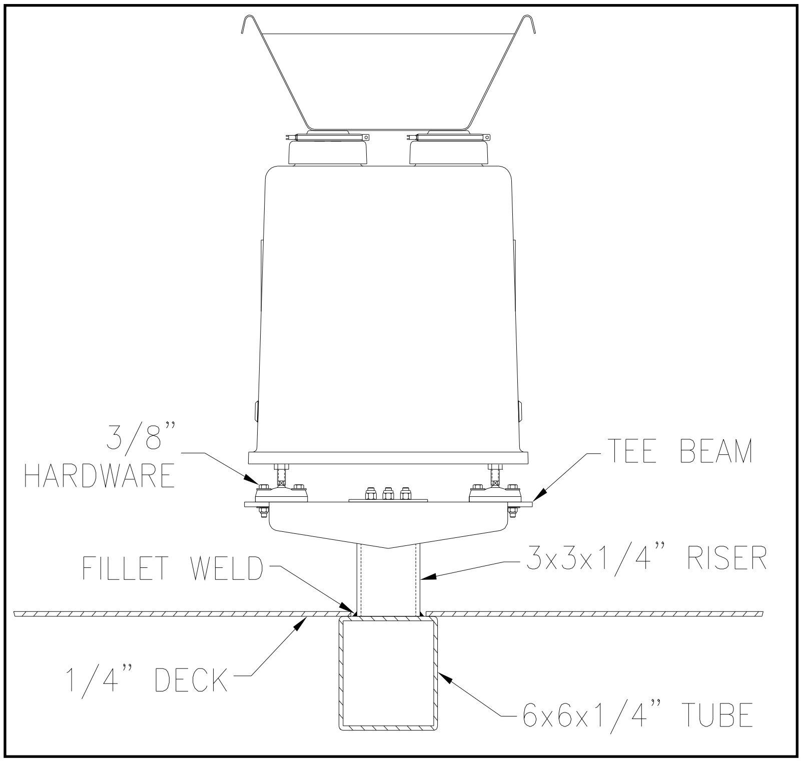

Figure 3-7 – T-Pedestal Welded to Central Spine

CAUTION

The dimensions above are minimum standards. Use of materials below these standards will cause increased vibration in the support structure, reducing efficiency and overall performance.

- Use 3/8-inch hardware to bolt the articulating feet of the FastBack® Models 90E or 90E-G2 to the deck support structure. Figure 3-8 shows a typical T-pedestal installation.

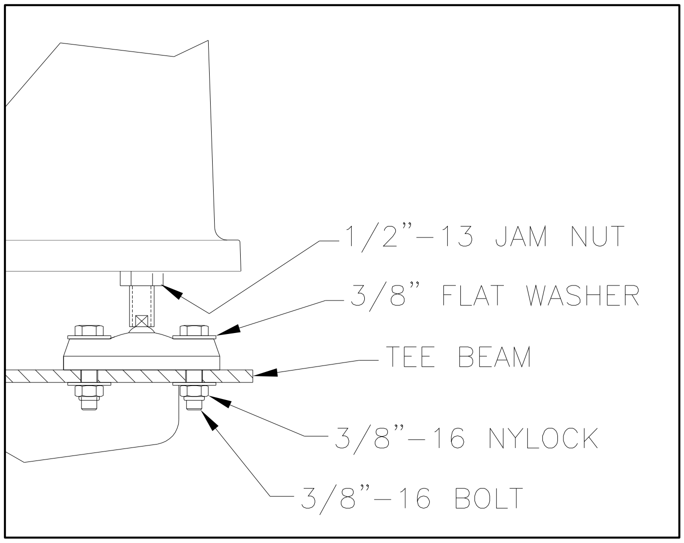

Figure 3-8 – Articulating Foot Hardware

- If the FastBack® Model 90E / 90E-G2 is to be placed directly on the deck, the articulating feet should be bolted through to a 4x4x¼-inch piece of angle iron which is in turn welded to the central spine described above (see Figure 3-9). Do not bolt the FastBack® to the deck alone.

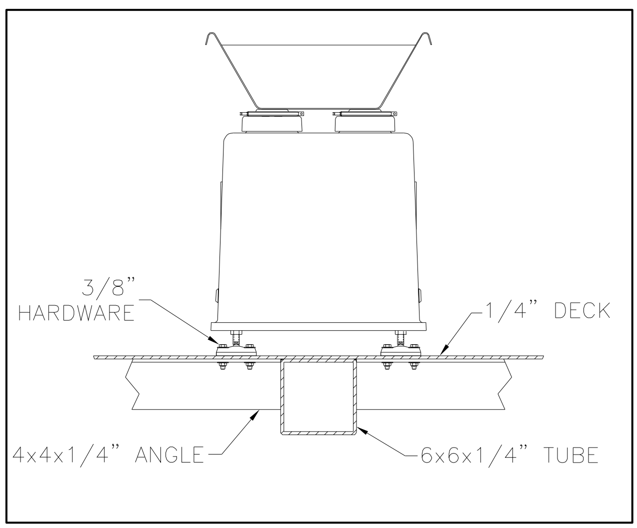

Figure 3-9 – Support Structure for Articulating Feet

CAUTION

The dimensions above are minimum standards. Use of materials below these standards will cause increased vibration in the support structure, reducing efficiency and overall performance.

Electrical Requirements

Table 3-3 shows the power and control voltage requirements for the FastBack® Models 90E and 90E-G2.

Primary Power (Nominal)	**Yaskawa Power Module**	***IFM efector* Photoeye**	**Hyde Park Ultrasonic Sensor**
3-phase 240 VAC 50-60 Hz 5-amp fusing	200-230 VAC 50-60 Hz 1.2 amp draw 5-amp fusing	Dual voltage 20-250 V AC/DC 2-wire sensor	10-24 VDC only 3-wire sensor
3-phase 480 VAC 50-60 Hz 5-amp fusing	380-460 VAC 50-60 Hz 0.6 amp draw 5-amp fusing		

Table 3-3 - 90E / 90E-G2 Power and Control Voltage Requirements

Do not run main power wires and control wires in the same conduit.

CAUTION

Always use flexible conduit to connect the FastBack® platform to the weigher platform. Do not use rigid conduit; this will transmit vibrations from the FastBack® to the weigher, which may decrease scale efficiency.